

Gamification in e-Learning Graduate Education: A Mixed-Methods Analysis of Its Impact on Academic Performance and Student Satisfaction

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Abstract— Graduate programs, known for their rigor and high academic demands, can generate significant levels of stress and demotivation among students. Therefore, it is important to implement active strategies that capture their attention, motivate them, and improve their academic performance. This study aims to investigate whether the implementation of a gamified learning approach can be related to academic performance, a topic with limited research at this level. To this end, a mixed methodology was used that collected data on students' final grades, as well as a Likert-scale satisfaction survey. Additionally, their perceptions and opinions about the strategy applied in class were gathered through an open-ended question questionnaire. Quantitative data were analyzed using a Spearman correlation, while qualitative data were worked through content analysis. The results indicated that, in the context of graduate studies, gamification is linked to improvements in both formative and summative assessments, with an average of 9.78/10. Moreover, a strong positive correlation ($r = 0.673$) was found between students' satisfaction with the gamification strategy and the grades they obtained throughout the course, with this result being statistically significant ($p < 0.05$). Consequently, this strategy can enhance the educational experience in these programs, increasing both satisfaction with the learning process and students' academic performance.

Keywords—gamification, educational innovation, Games, e-learning, Graduate education, gamified learning.

I. INTRODUCTION

E-learning education has revolutionized the way formal education is perceived, providing an alternative to face-to-face education where time and location are constraints when accessing knowledge. According to [1], this teaching modality represents an evolution from distance, hybrid, and in-person education, as technology serves as a mediator between students and the knowledge to be acquired, facilitating lifelong learning. In this regard, technological and pedagogical resources provide support to offer quality educational programs that create new learning spaces.

According to [2], these e-learning spaces are online environments where students are guided through tutorials during their learning experience. To achieve this, materials, activities,

communication tools, and assessment procedures must be provided to develop autonomy and self-regulation in students, as the role of the teacher is to act as a tutor. Similarly, [3] mentions that the sociocultural nature of collaborative work is developed from any location where participants have an adequate internet connection, enhancing knowledge construction in a learning community.

However, Chaw & Tang [4] emphasize that regardless of the teaching modality, efforts must be made to generate student commitment and motivation to complete class activities and tasks. This is particularly crucial in online teaching and learning processes, where students undertake many of the activities independently. Furthermore, due to individuals having different learning styles, diverse work strategies should be sought.

In terms of online graduate education, significant challenges related to learning satisfaction and its reflection on grades are faced. Graduate programs, characterized by rigor and high academic demands, can lead to high levels of stress and demotivation. These students face multiple professional and personal demands, which can affect their commitment to learning. Moreover, as programs often include experienced professionals, they seek self-directed, relevant, and practical learning that is adapted to reality and contextualized, using methods and strategies that value their autonomy and previous experience [5].

Under this need, educational innovation seeks to create more effective, motivating, and meaningful learning experiences for students, preparing them for real-life situations in their contexts. In this regard, gamification as a pedagogical strategy offers a promising approach to transform learning and enhance the academic performance of students.

Gamification involves using game elements and adapting them to non-game contexts [6], [7]. In educational settings, it aims to generate active and motivating learning that contributes to problem-solving [8]. Moreover, this strategy seeks to improve students' academic performance through a playful experience, involving a pedagogical design that goes beyond simply introducing games in the classroom [9].

In this context, according to [10], gamification can be either superficial or deep. The former includes ludic elements to motivate students through extrinsic things or situations like rewards or prizes. In contrast, the latter focuses on activities that are inherently interesting and relevant to students, fostering intrinsic motivation. To develop this type of gamification, [11] proposes the presence of four vectors known as RAMP: "Relatedness," the desire to establish links and connections with others; "Autonomy," the need to feel independence and control over the learning process; "Mastery," the desire to acquire new skills and develop expertise; and "Purpose," the search for meaning and transcendence in our actions.

This strategy is also supported by motivational theories such as Self-Determination Theory [12], which posits that individuals have three basic psychological needs: competence, autonomy, and relatedness. Gamification can satisfy these needs by providing a learning environment that is challenging yet achievable (competence), allows students to make decisions (autonomy), and facilitates social interaction (relatedness). The use of gamification elements, such as point systems, rewards, and leaderboards, can increase students' intrinsic motivation by making educational tasks more engaging and rewarding [12]. Baena and García [13] point out that gamification can transform routine activities into exciting learning experiences, promoting greater engagement and perseverance among students.

Besides, in the context of gamification, Flow Theory [14] is fundamental because many elements of games, such as progressive challenges, immediate feedback, and clear goals, are designed to induce this state of flow in participants. By applying these elements in educational settings, gamification can help students enter a state of flow, which enhances their focus, motivation, and enjoyment of the learning process, thereby contributing to more effective and meaningful learning. Therefore, Flow Theory complements Self-Determination Theory by providing a deeper understanding of how gamification can create learning experiences that not only meet basic psychological needs but also keep students engaged and motivated through a balance between challenge and skill.

Gamification consists of dynamics, mechanics, and game components. Dynamics constitute the world created for users to use the gamified system. These dynamics initiate mechanics, thereby determining student behavior. They are related to student motivation and are based on constraints, emotions, narrative, progression, and relationships. Mechanics are actions created to engage the user in the system and their overall progression. Their real purpose is to materialize what has been defined in the game dynamics. Game components are the features associated with games that provide feedback to players, motivate them to perform mechanics, and help drive the dynamics present in the system [15].

Some key components of gamification include (i) avatars, which are visual representations that participants create or select to identify themselves, fostering emotional attachment and expressing their originality and personality. (ii) Tutorials, designed to familiarize students with the activity, explain rules, and provide strategies for optimal performance. (iii) Missions, which are tasks with clear objectives and diverse solution alternatives, starting with simple and progressive activities that

allow for benefits. (iv) Points, as numerical units, represent progress and determine victory or rankings, while leaderboards organize participants' performance, increasing competitiveness and motivation to achieve top positions [16].

TABLE I. TAXONOMY OF THE MOST COMMON GAME ELEMENTS IN EDUCATION

Game element	Concept	Motivation	Dimension
Reputation	Titles, rankings, or statuses that a student can obtain, earn, and accumulate. Represents a social status that does not necessarily reflect players' skills. Used to create a hierarchy within the game environment or communities.	Intrinsic	Social
Mission	Provides students with a goal or purpose for completing tasks, such as receiving discounts after achieving a certain score on an assignment. Also known as quests, side quests, pending tasks, milestones, or objectives.	Intrinsic	Particular
Avatar	A virtual identity is chosen by the students to represent themselves in the gamification. In some cases, they can customize it with real or fictional physical appearances.	Intrinsic	Immersion
Rules	Guidelines that tell players what is and is not allowed in a particular situation.	Extrinsic	Context
Sharing	The action is taken when students share content with other students, groups, or specific individuals. They enjoy sharing content with others, which can be done on social media or other electronic means.	Extrinsic	Media and Communication
Reward-Prize	A positive consequence for a student is a result of a desirable behavior or action, such as earning extra points for punctuality.	Extrinsic	Achievement

Moreover, in gamification, notable mechanics include (a) competition, which introduces a dynamic of winning and losing, although it is recommended to foster personal competition to avoid demotivation; (b) feedback, which facilitates students' self-reflection by analyzing their learning from immediate results; and (c) challenges, which involve effort to overcome a difficult task and must be balanced to maintain motivation. (d) Rewards, such as points, medals, or advancing to higher levels, are obtained by overcoming challenges and can encourage additional effort in users. Finally, (e) the shifts encourage equal and sequential participation of students, promoting cooperation and teamwork [17], [18]. According to [16], they propose a taxonomy of the most common elements within gamification, including their definition, concepts, type of motivation provoked, and dimension (see Table I).

Similarly [16] denominate the work dimensions as follows in Table II:

TABLE II. DIMENSIONS OF GAMIFICATION

Dimension	Description
Social	Related to the interactions between students and the game environment. Without social elements, students may feel isolated or unable to interact with others. Examples include reputation, competition, or team cooperation.
Achievement	Reflects the student's progress and can be used to provide feedback. The absence of this dimension may leave students feeling lost due to a lack of clear system responses to their actions. Examples include points or levels.
Particular	Refers to the learner's personal experience with the game environment. The absence of these elements may lead to demotivation as the service does not adapt to them. Examples include missions or user profiles.
Immersion	Reveals the fictitious habitat or space where the game unfolds through storytelling that connects the student's experiences with the context. Fictional elements can provide context or create an immersive experience. Examples include player roles, narratives, or avatars.
Context	Related to the environment in which gamification is implemented. Game elements can be represented as properties. Without these elements, the game environment may appear boring. Examples include economy or lifespan.
Media and Communication	Related to student interactions with social media and other technologies for chatting, sharing, posting, writing, etc. These elements allow students to express feelings, thoughts, votes, etc., to teachers or others. Examples include posting or location tagging.

To develop a deep gamified experience and make the most of its dynamics, mechanics, and elements, there are several framework designs. For this work, the "Six Steps to Gamification" model is considered (see Fig. 1), which is based on the following steps: (1) defining the goals to be achieved; (2) outlining the target behaviors expected from users; (3) describing the player profiles (interests, what motivates them); (4) designing activity loops (the process users should follow);

(5) planning how to keep users returning; and (6) implementing the appropriate tools (how interaction will be measured, scoring systems, badge allocation, etc.) [15].

The effective integration of these motivational elements and enablers into a gamified system can significantly enhance user engagement, satisfaction, and performance. By addressing core psychological needs through thoughtful game design, educators and designers can create environments where participants are not only motivated to engage but also driven to excel, thereby achieving both personal and educational goals.

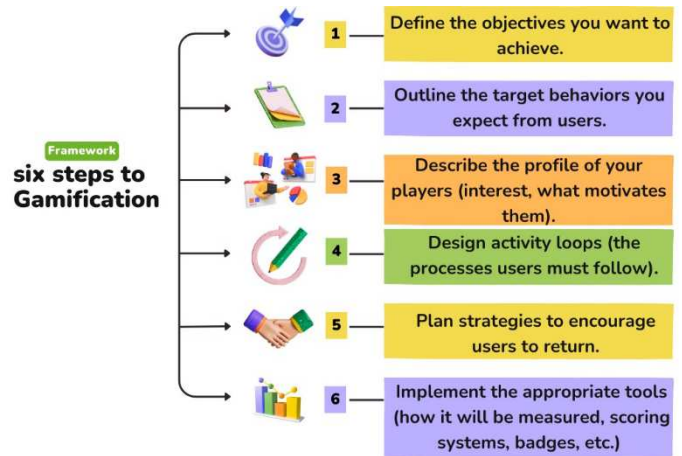


Fig. 1. Six steps to Gamification adapted from [15].

In the context of higher education, Sailer and Homner [19] demonstrated that gamification can transform the way students engage with and interact with class content, promoting greater engagement and satisfaction. However, specific research on the application of this strategy at the graduate level is limited, making this study particularly novel. This work focuses on graduate students, a relatively underexplored population in gamification research. Through a detailed and empirical analysis, this study aims to provide robust evidence on the association between gamification and academic performance in graduate education. By doing so, it seeks to contribute significantly to the existing body of knowledge and offer practical insights into implementing gamification strategies in graduate programs. The goal is to enhance the educational experience and improve academic outcomes for students at this advanced level of education.

II. METHODOLOGY

The present study was developed in a Higher Education Institution in Ecuador in the area of online graduate studies. The class that was part of the research process is called "Gamification: Education Through Play" and belongs to the Master's program in Pedagogy, with a focus on Teaching and Educational Innovation. This class implemented a gamification process based on the Six Steps to Gamification framework by Werbach and Hunter [15], which was carried out as follows:

1. Learning Objectives: Understand the theoretical foundations of gamification; design and apply gamification strategies in the classroom.

2. Target Student Behaviors: The aim is for students to develop teamwork skills, autonomy, research abilities, and problem-solving capabilities.

3. Student Profile: This course is attended by 55 students, aged between 35 and 55, of which 42 are women and 13 are men. Among the attendees, 80% (44) hold bachelor's degrees related to education, while 20% (11) have degrees in other fields. The students are motivated to learn practical activities that they can adapt to their educational contexts.

4. Designed Activities: The tool Classcraft - Relationships are everything [20] was used to gamify this class. This tool is well-known and utilized by various educational communities in different countries, allowing for a virtual classroom setting where the learning progress of each student can be tracked. As a gamified space, it allows configurations for avatars, rules, scores, leaderboards, and customized activities. The following activities were conducted in this space (Fig. 2):

- **Introduction to the Narrative:** Students were invited to participate in a series of activities where they would leave their student role behind and take on a role within a provided narrative. In this case, the students found themselves in a post-apocalyptic world with only one Supreme Master who dictated the order and life on the planet. The surviving students have been divided into 11 tribes scattered throughout the planet. Each of these tribes has members with special powers granted by the Supreme Master. Thanks to these powers, members can be warriors, healers, or wizards. They have been summoned to reintroduce education into the world. To achieve this, each tribe will embark on research about gamification, creating an education that is motivating and necessary for the new times.
- **Introduction to the Workspace:** When students accessed Classcraft, they were assigned one of the mentioned characters. Similarly, the platform explained which actions would help gain points and which would diminish them. Actions such as completing a task early or on time, providing additional information, or going beyond what is requested could increase their life points; however, actions such as being rude, not working in a team, or not submitting work on time could decrease their points.
- **Work Activities:** Within the given context, teachers had to work with their "tribe" to create a classroom plan centered on gamified processes for their students. This planning was conducted over eight weeks, with synchronous meetings once a week. During these meetings, tribes were provided with theory, resources, examples, and feedback on their progress.
- **Evaluation:** To monitor the training process, both formative and summative assessments were conducted. At the end of each unit, students had to complete a questionnaire on the topic, and throughout the formative process, they were progressively accompanied in their planning.

5. Actions and selection of tools to motivate students: To motivate and engage students, several game elements were used, the same are detailed in Table III.

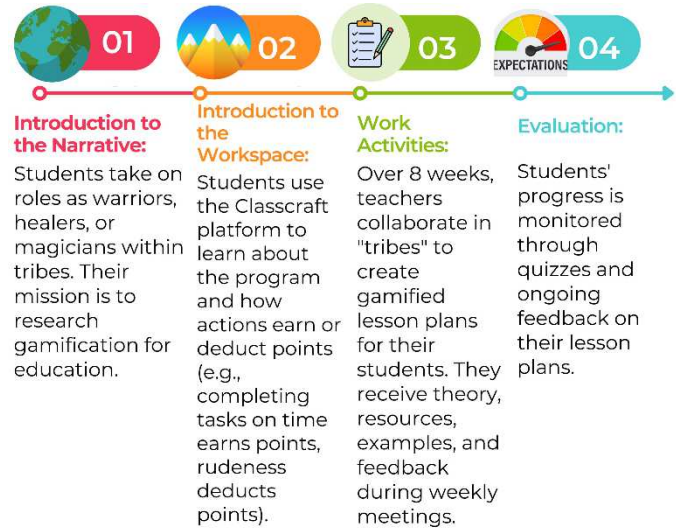


Fig. 2. Designed activities in the Classcraft world.

TABLE III. ELEMENT PLANNING, MOTIVATION, AND DIMENSION

Game Element	In the Planning	Motivation	Dimension
Competition	Each team aims to finish the task early or add a bonus to the given instruction.	Intrinsic	Social
Cooperation	Teams work together to achieve the same goal.		
Social Interaction	Each team shares the activity with the class to exchange ideas.		
Progress Bar	Students can see their accumulated points and their level on the platform.	Extrinsic	Achievement
Reward-Prize	Students receive points for performing correct actions in class.		
Level	As students accumulate points, they can level up.		
Badge	For each level obtained, students receive a badge.		
Points	Points are added or deducted for each action.	Intrinsic	Particular
Mission	From the general narrative, missions are divided for teams to complete.		
Objective	Teams work together to create a plan for their educational contexts.		
Personalization	The themes of the planning were		

	contextualized to each student's educational community.		
Emotion	Videos, audio, and escape rooms were used for the experience.		
Roles	Within each team, members assumed a role to achieve the activity's objective.	Intrinsic	Immersion
Narration	All activities are developed within the narrative of a post-apocalyptic world.		
Avatar	Each student could be a warrior, healer, or mage.		
Rules	Students can earn points for finishing tasks early, helping a classmate, spotting errors, providing additional information, or using innovative tools. They can lose points for not completing tasks, being rude, or submitting late.	Extrinsic	Context
Time	Students had 8 weeks to complete the activity, with a synchronous meeting of two hours per week.		
Chat	Students could share their doubts, comments, or suggestions in the class group chat.	Extrinsic	Media and Communication
Tagged Locations	Each mission was tagged in a world within Classcraft.		

Furthermore, to measure the impact of this strategy, a mixed-methods inferential approach was employed. The dependent variable was the grades obtained by students throughout the course, while the independent variable was the satisfaction with the use of gamification in their academic training. Given that this educational aspect aims to generate initiatives considering the educational community, a design-based research method was used, following the model proposed by Benito & Salinas [21], based on the Reeves model (see Fig. 3). It is ideal for education because it allows for continuous iteration and improvement of educational interventions in real-world contexts. This approach integrates theory with practice, ensuring that solutions are relevant and effective for the specific problems faced by educators and students. For this purpose, participant observation, surveys, and tests were used as data collection techniques.

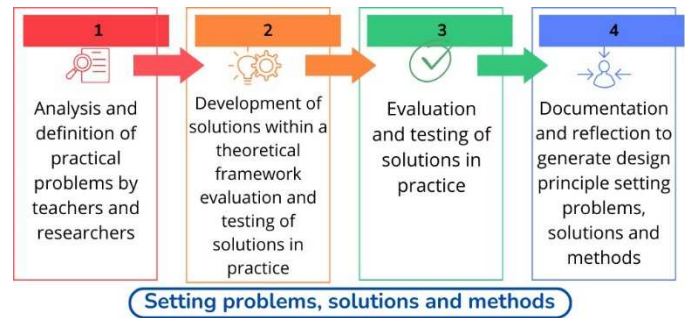


Fig. 3. Phases of research based on Reeves's design [21].

The research process included the following phases:

- Phase One: This phase involved a review of the literature combined with the identification of the problem and possible practical solutions.
- Phase Two: The gamified lesson was planned, considering the interests and needs of the students.
- Phase Three: The process and the students were evaluated through tests and the progressive results of the proposed activities. In addition, information was collected on their satisfaction with the learning experience.

This structured approach allowed for a comprehensive analysis of the effectiveness and impact of gamification on students' academic performance and satisfaction.

III. ANALYSIS AND RESULTS

The quantitative data included in the analysis were collected through a satisfaction survey at the end of the course. This survey evaluated the training experience using a Likert-type scale, where the highest score was five and the lowest was one. One of the items in the survey was "Evaluation of your experience in the virtual environment". Additionally, the final grades of the course were used as part of the analysis. This process yielded both ordinal and continuous data, which were the result of various practical activities that the students completed in each subject.

Regarding the grades, it was observed that within the group, there was a mean of (9.78 ± 0.35) . In this context, the minimum value was 8.45 and the maximum value was 10. Placing the participants in the category of excellence in their training process. Furthermore, both types of data were analyzed using a Spearman correlation test in the IBM SPSS® software, since the Kolmogorov-Smirnov test ($n > 50$) for normality indicated that the data did not follow a normal distribution ($p < 0.05$), and there was the presence of ordinal and continuous variables. From this, a high positive correlation ($r = 0.673$) was found between the satisfaction with the use of the gamification strategy in their academic training and the grades obtained throughout the course by the students, with this test being significant ($p < 0.05$) (see Table IV).

TABLE IV. SPEARMAN CORRELATION RESULTS

Correlations				
			Grade Report	Use Satisfaction
Spearman's Rho	Grade Report	Correlation Coefficient	1.000	0.673**
		Sig. (two-tailed)	.	0.000
		N	55	55
	Gamification Use Satisfaction	Correlation Coefficient	0.673**	1.000
		Sig. (two-tailed)	0.000	.
		N	55	55

**The correlation is significant at the 0.01 level (two-tailed)

On the other hand, the qualitative data were collected through an open-ended questionnaire that gathered students' perceptions and opinions about the methodology used in the class at the end of the course; the question used was "How satisfied are you with the Gamification class in terms of content, methodology, and relevance to your learning? Please detail your experience and provide suggestions for future improvements". Table V presents the data analyzed through content analysis, revealing that students value aspects of competence, knowledge, pedagogical strategy, and learning environment developed by the teacher.

TABLE V. OPEN-ENDED QUESTIONNAIRE THAT COLLECTED THE STUDENTS' PERCEPTIONS AND OPINIONS ON THE METHODOLOGY USED IN THE CLASS.

Category	Description	Example
Teacher competence and knowledge	Students recognized the background and form of the teacher to teach the subject.	Very interesting classes, with his mastery of the subject and explanation has been learning every day. Thank you for your patience and for sharing your knowledge.
Teaching strategies	Students valued the strategy used, the use of various resources, and the contextualization of activities.	A very enriching didactic strategy where we can link several activities in a positive, fun, and stimulating way.
Learning environment	The students appreciated that they had a pleasant space, conducive to learning and active participation.	I was fascinated and very entertained. I was able to participate and exchange ideas with my tribe. Thank you very much for your great help in your teaching work.

IV. DISCUSSION AND CONCLUSION

In the current educational environment, the search for innovative strategies to enhance learning and, consequently,

students' grades, is constant. In this sense, gamification has emerged as a tool within educational innovation with the potential to transform the way students interact with knowledge and, consequently, positively impact their academic performance. According to Deterding [22], gamification can improve academic performance by providing immediate and structured feedback, which helps students identify and correct their mistakes promptly.

This strategy also encourages more active and participatory learning, which are essential elements for superior academic performance, as students value the quality of teaching, highlighting aspects such as teachers' competencies, strategies, and attention received. According to Landers [23], it is crucial that the game elements are integrated coherently with the educational objectives and that they are adapted to the students' levels of competence and experience. Similarly, in the research by Urh et al., [24] they mention that the main phases of development are: analysis, planning, development, implementation, and evaluation. Throughout all the phases of e-learning development, elements of user experience are very important.

In conclusion, gamification, as a pedagogical strategy, has the potential to transform the educational experience in graduate programs, improving satisfaction with the learning experience and students' academic performance. Furthermore, as indicated by [25], deploying a greater number of gamification and engagement elements has been shown to positively impact student engagement, which can lead to constructive outcomes for the effectiveness of e-learning.

However, to maximize these benefits, careful design and implementation that considers the unique characteristics of graduate students is essential. By introducing new learning experiences to learners, gamification can become a catalyst for student academic success, aligning with the principles of current educational innovation.

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